

Quiz 2 - Answer Key

Thursday, March 5, 2026 8:51 AM

Question 1:

Which scheduling algorithm assigns each process a fixed time slice (quantum) and cycles through them?

- ☐ First Come, First Serve (FCFS)
- ☐ Shortest Job Next (SJN)
- ☒ Round Robin
- ☐ Priority Scheduling

Question 2:

What information is saved during a context switch?

- ☒ The process' CPU registers and program counter
- ☐ The entire contents of RAM
- ☐ Only the process ID
- ☐ The process' priority level

Question 3:

Which of the following best describes *readylist* in the Xinu operating system?

- ☐ A queue that holds terminated processes
- ☒ A priority queue that stores processes waiting for CPU execution
- ☐ A list of blocked I/O operations
- ☐ A shared memory buffer for interprocess communication

Question 4:

A race condition occurs when:

- ☐ A process runs in user mode instead of kernel mode
- ☒ Multiple processes access shared resources without synchronization
- ☐ The scheduler runs a high-priority process before a low-priority one
- ☐ A thread exceeds its time quantum

Question 5:

Which of the following is a drawback of First-Come, First-Serves (FCFS) scheduling?

- ☐ It requires complex hardware support
- ☒ It can lead to convoy effect, where short processes wait behind long ones
- ☐ It does not use CPU time efficiently
- ☐ It always executes processes in the shortest time possible

Question 6:

What happens when the *resched* function in Xinu is called? Discuss specifically how the *prprio* and *prstate* variables on the *procent* structure (part of *proctab[]*) are used.

Answer:

Question 7:

After profiling the performance of the Xinu scheduler, you find that high priority processes are starving low priority processes. This is leading to a very high turnaround time and unfairness. How can the scheduler algorithm be changed to improve turnaround time and fairness?

Answer: